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# Uncertainty Quantification for Scientific Machine Learning using Sparse Variational Gaussian Process Kolmogorov-Arnold Networks (SVGP KAN)

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## Abstract

Kolmogorov-Arnold Networks have emerged as interpretable alternatives to traditional multi-layer perceptrons. However, standard implementations lack principled uncertainty quantification capabilities essential for many scientific applications. We present a framework integrating sparse variational Gaussian process inference with the Kolmogorov-Arnold topology, enabling scalable Bayesian inference with computational complexity that is in theory quasi-linear in sample size. Through analytic moment matching, we propagate uncertainty through deep additive structures while maintaining interpretability. We use three example studies to examine the framework’s ability to distinguish aleatoric from epistemic uncertainty: calibration of heteroscedastic measurement noise in scalar field prediction, quantification of prediction confidence degradation in multi-step forecasting of advection-diffusion dynamics, and out-of-distribution detection in convolutional autoencoders. These results suggest Sparse Variational Gaussian Process Kolmogorov-Arnold Networks (SVGP-KANs) as a promising architecture for uncertainty-aware learning in scientific machine learning.

## 1. Introduction

The quantification of uncertainty in neural network predictions is an important requirement in scientific machine learning, medical diagnosis, autonomous systems, and engineering design. Traditional neural architectures based on multi-layer perceptrons provide point estimates without principled measures of confidence. This limits their deployment in applications where understanding model uncertainty is just as critical as prediction accuracy. While Bayesian neural networks offer a theoretical framework for uncertainty quantification, their computational demands and challenges in posterior inference have motivated the development of alternative approaches combining expressivity with tractable probabilistic reasoning.

The Kolmogorov-Arnold network [1,2] represents a major shift in neural architecture design by leveraging the Kolmogorov-Arnold representation theorem. This theorem establishes that any multivariate continuous function defined on a finite domain can be represented as a finite composition of continuous univariate functions and addition operations. Unlike conventional multi-layer perceptrons that employ fixed nonlinear activations at nodes, Kolmogorov-Arnold networks place learnable univariate functions on edges, creating an additive structure that enhances interpretability. The original implementation employs B-spline parameterizations for these univariate functions, enabling direct visualization and symbolic regression capabilities that have proven valuable in scientific discovery tasks.

Recent extensions have explored various formulations and applications of Kolmogorov-Arnold networks across diverse domains as detailed in recent comprehensive review articles [3,4]. For instance, Bodner et al. [5] developed convolutional variants, demonstrating that the edge-based learning paradigm can be successfully applied to computer vision tasks. Zhang and Zhang [6] introduced graph-theory extensions for relational data, while Genet and Inzirillo [7] and Yang and Wang [8] proposed temporal adaptations for sequence modeling. Alternative basis functions also have been investigated. Xu et al. [9] replaced B-splines with Fourier bases for improved computational efficiency, Yang and Wang [8] explored rational functions to reduce parameter counts, Bozorgasl and Chen [10] employed wavelet decompositions for multi-resolution analysis, and Li [11] demonstrated equivalences between certain Kolmogorov-Arnold formulations and radial basis function networks. Zheng et al. [12] analyzed the theoretical properties of spline knots in B-spline-based implementations, deriving tight upper bounds on representational capacity and proposing free-knot adaptations to enhance expressive power while maintaining training stability.

Despite these advances, the deterministic nature of standard Kolmogorov-Arnold networks limits their applicability in uncertainty-critical domains. Hassan et al. [13] attempted to address this limitation through Bayesian Kolmogorov-Arnold networks. They placed Gaussian priors on individual B-spline coefficients, yielding weight-space posteriors. The learned distribution quantifies uncertainty in the coefficients themselves, not in the function they parameterize, and the resulting credible bands depend on the choice of basis discretization rather than reflecting any function-space prior. That is, while their approach captures weight-space uncertainty, it does not fully capture the function-space uncertainty inherent in non-parametric Bayesian modeling.

Another previous study [14] avoided this limitation by modelling edge functions directly as Gaussian process priors, achieving principled function-space uncertainty that is independent of any fixed parameterization. Although this formulation provides these theoretical benefits, it relies on exact inference and requires inversion of  $N \times N$  covariance matrices with  $O(N^3)$  computational complexity, limiting scalability. The present framework preserves the function-space perspective of [14] while replacing exact inference with sparse variational approximations [15,16]: inducing point representations reduce per-sample complexity from  $O(N^3)$  to  $O(NM^2)$  and with  $M \ll N$ , enabling training on

datasets orders of magnitude larger. Additionally, the KAN additive topology permits independent variational posteriors per edge, avoiding the factorial inter-layer coupling that would arise in a general deep GP architecture.

Beyond architectures derived from Kolmogorov-Arnold networks, the broader scientific machine learning community has developed several approaches for uncertainty quantification in neural network predictions, which may be organized by where stochasticity is introduced. Weight-space Bayesian methods place prior distributions on network parameters and propagate uncertainty through the network mapping; examples include variational Bayes formulations for operator networks [17] and Monte Carlo dropout [18], which repurposes dropout regularization as approximate Bayesian inference through multiple stochastic forward passes. However, weight-space priors such as  $\mathcal{N}(0, \mathbf{I})$  on parameters often lack physically interpretable meaning [19]. Ensemble methods [20] capture uncertainty through multiple independently trained models, providing effective uncertainty estimates but at proportional increases in training and memory cost; recent variants aim to improve efficiency through snapshot training [21] or batch parameterization [22]. In scientific computing specifically, physics-informed neural networks have been augmented with probabilistic components [23,24], though these typically address either aleatoric or epistemic uncertainty in isolation. An alternative paradigm places priors directly on the functions being learned rather than on network parameters. Sparse Gaussian process approximations [15,16] offer function-space uncertainty with favorable scaling properties, and deep Gaussian processes [25] extend this to hierarchical compositions, though at the cost of intractable inter-layer dependencies. Functional variational methods [26] bridge weight-space and function-space perspectives by directly parameterizing function-space posteriors, while moment-based approaches [27] achieve single-forward-pass variance computation through analytic moment propagation. Zou and Yan [28] recently proposed probabilistic formulations using sparsified deep Gaussian processes with additive kernels, though their focus remained on theoretical properties rather than practical applications requiring fine-grained spatial uncertainty quantification.

We introduce a framework that adopts the function-space perspective by embedding sparse variational Gaussian processes within the structured additive topology of Kolmogorov-Arnold networks. This framework can in theory achieve quasi-linear scalability while preserving interpretability and uncertainty quantification capabilities. The three advantages are: GP kernel hyperparameters (length-scales, signal variances) encode physically interpretable prior information; the univariate edge-function structure avoids the inter-layer coupling that complicates general deep GP inference; and the orthogonal variance decomposition provides geometric guarantees that predictive uncertainty reverts to the prior for out-of-distribution inputs.

Our approach employs inducing point approximations following [15] and [16], reducing per-sample complexity from  $O(N^3)$  to  $O(NM^2)$  where  $M$  represents the number of inducing points. The univariate structure of Kolmogorov-Arnold edges enables independent variational approximations across edge functions, avoiding the computational challenges associated with high-dimensional variational inference in general deep Gaussian

processes. Through analytic moment matching [29], we propagate input uncertainty through layers via closed-form expressions, maintaining computational tractability while quantifying both aleatoric uncertainty arising from measurement noise and epistemic uncertainty reflecting model limitations.

The remainder of the manuscript is organized as follows. First, we discuss a mathematical framework for sparse variational Gaussian process inference adapted to Kolmogorov-Arnold topology, presenting tractable evidence lower bounds and gradient expressions for stochastic optimization. Next, we present application of the framework to study three different example problems with relevance to fluid mechanics, scalar transport, and anomaly detection in biomedicine and other disciplines. In the first example study, we demonstrate calibration of spatially varying aleatoric uncertainty in a transport problem with heteroscedastic measurement noise. In the second example study, we quantify epistemic uncertainty growth in multi-step forecasting of advection-diffusion transport, showing that prediction confidence naturally degrades with forecast horizon due to the accumulation of model approximation errors in sequential predictions. In the third example study, we establish out-of-distribution detection capabilities through integration with convolutional autoencoders.

## 2. Mathematical Framework

### 2.1 Kolmogorov-Arnold Networks and Probabilistic Edge Functions

A Kolmogorov-Arnold network (KAN) transforms an input vector  $\mathbf{x} \in \mathbb{R}^{D_{\text{in}}}$  to an output  $\mathbf{y} \in \mathbb{R}^{D_{\text{out}}}$  through a composition of layers, where each layer implements an additive aggregation of univariate transformations. For a single layer mapping dimension  $D_{\text{in}}$  to dimension  $D_{\text{out}}$ , the  $j$ -th output component is

$$y_j = \sum_{i=1}^{D_{\text{in}}} \phi_{j,i}(x_i) + b_j$$

where  $\phi_{j,i}: \mathbb{R} \rightarrow \mathbb{R}$  denotes a univariate function associated with the edge connecting input dimension  $i$  to output dimension  $j$ , and  $b_j$  represents an optional bias term. This formulation differs fundamentally from conventional multi-layer perceptrons, which apply fixed activation functions after affine transformations. The Kolmogorov-Arnold structure places learnable nonlinearities on edges rather than nodes, enabling direct interpretation of individual feature contributions through visualization of univariate functions.

In deterministic implementations,  $\phi_{j,i}$  are parameterized using, for example, B-splines, Chebyshev polynomials, Fourier bases, or rational functions. We instead model each edge function as a Gaussian process, introducing a probabilistic interpretation that naturally quantifies uncertainty. Specifically, we place a zero-mean Gaussian process prior over each  $\phi_{j,i}$ :

$$\phi_{j,i} \sim \mathcal{GP}(0, k(\cdot, \cdot; \theta_{j,i}))$$

where  $k: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  denotes a kernel function parameterized by hyperparameters  $\theta_{j,i}$  that may be learned during training. This construction transforms the deterministic Kolmogorov-Arnold layer into a probabilistic mapping where the output  $y_j$  becomes a random variable with uncertainty quantified through its posterior distribution.

The choice of kernel function determines the inductive bias of the model. For universal approximation with smooth functions, we primarily employ the squared exponential (radial basis function) kernel:

$$k_{\text{SE}}(x, x'; \sigma_f^2, \ell) = \sigma_f^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right)$$

where  $\sigma_f^2$  controls the prior variance of function values and  $\ell$  represents the lengthscale determining the correlation structure. Alternative kernels such as the Matérn family may be employed when modeling phenomena with specific smoothness properties or when incorporating prior knowledge about function regularity. However, the closed-form KL divergence calculations presented in Section 2.2 are specific to the RBF kernel; extensions to other kernel families require deriving the corresponding KL expressions.

## 2.2 Sparse Variational Inference for Scalable Training

Exact Gaussian process inference requires computing and inverting the  $N \times N$  Gram matrix  $\mathbf{K}$  whose elements are  $K_{nm} = k(x_n, x_m)$  for training inputs  $\{x_n\}_{n=1}^N$ . The computational complexity of  $O(N^3)$  for matrix inversion and  $O(N^2)$  storage requirements prohibit application to large datasets. We employ sparse variational approximations to reduce this complexity while maintaining the theoretical benefits of Bayesian inference.

For each edge function  $\phi_{j,i}$ , we introduce a set of  $M$  inducing inputs  $\mathbf{Z}_{j,i} = \{z_m\}_{m=1}^M$  where typically  $M \ll N$ . The inducing inputs represent pseudo-datapoints in the input space, and their corresponding function values  $\mathbf{u}_{j,i} = \phi_{j,i}(\mathbf{Z}_{j,i})$  are termed inducing variables. Following [15], the key insight is that conditioning on  $\mathbf{u}_{j,i}$  renders function values at different data points approximately independent:

$$p(\phi_{j,i}(\mathbf{X}) \mid \mathbf{u}_{j,i}) \approx \prod_{n=1}^N p(\phi_{j,i}(x_n) \mid \mathbf{u}_{j,i})$$

This factorization reduces the computational burden from  $O(N^3)$  to  $O(NM^2)$ , enabling application to large datasets.

We employ the variational framework [16] to approximate the true posterior  $p(\mathbf{u}_{j,i} \mid \mathbf{y})$  with a parametric variational distribution  $q(\mathbf{u}_{j,i}) = \mathcal{N}(\mathbf{m}_{j,i}, \mathbf{S}_{j,i})$  where  $\mathbf{m}_{j,i} \in \mathbb{R}^M$  and  $\mathbf{S}_{j,i} \in \mathbb{R}^{M \times M}$  are variational parameters to be optimized. In our implementation, we further restrict  $\mathbf{S}_{j,i}$  to a diagonal matrix, reducing the variational parameters per edge from  $O(M^2)$

to  $O(M)$  while retaining per-inducing-point variance estimates. The variational posterior for the function value at an arbitrary input  $x$  is obtained through the conditioning equations:

$$q(\phi_{j,i}(x)) = \int p(\phi_{j,i}(x) | \mathbf{u}_{j,i}) q(\mathbf{u}_{j,i}) d\mathbf{u}_{j,i}$$

yielding a Gaussian distribution with mean and variance:

$$\begin{aligned} \mu_{j,i}(x) &= \mathbf{k}_x^\top \mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{m}_{j,i} \\ \sigma_{j,i}^2(x) &= k(x, x) - \mathbf{k}_x^\top \mathbf{K}_{\mathbf{ZZ}}^{-1} (\mathbf{K}_{\mathbf{ZZ}} - \mathbf{S}_{j,i}) \mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{k}_x \end{aligned}$$

where  $\mathbf{k}_x = [k(x, z_1), \dots, k(x, z_M)]^\top$  and  $\mathbf{K}_{\mathbf{ZZ}}$  denotes the  $M \times M$  kernel matrix evaluated at inducing inputs.

The evidence lower bound (ELBO) provides a tractable objective for stochastic optimization. For a single edge function with scalar outputs, the ELBO decomposes as:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(\phi)} [\log p(\mathbf{y} | \phi)] - \lambda \cdot \text{KL}[q(\mathbf{u}) \parallel p(\mathbf{u})]$$

where  $\lambda$  is a hyperparameter controlling the relative weighting of the KL divergence term. The KL divergence term admits a closed form for Gaussian distributions:

$$\text{KL}[q(\mathbf{u}) \parallel p(\mathbf{u})] = \frac{1}{2} \left( \text{tr}(\mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{S}) + \mathbf{m}^\top \mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{m} - M + \log \frac{|\mathbf{K}_{\mathbf{ZZ}}|}{|\mathbf{S}|} \right)$$

The KL weight  $\lambda$  balances data fidelity against prior regularization. Empirically,  $\lambda \in [0.001, 0.1]$  was observed to perform reasonably well across applications [30–33], with smaller values prioritizing data fit and larger values encouraging conservative uncertainty estimates. In our experiments, we employ  $\lambda$  of the order of 0.01 to provide a balanced trade-off between fitting capacity and regularization.

The expected log-likelihood term needs to be approximated numerically but can be efficiently estimated using mini-batches, enabling stochastic gradient descent optimization. The mini-batch approximation ensures that per-iteration complexity scales as  $O(BM^2N)$  where  $B$  denotes batch size, achieving in theory linear scaling in the full dataset size  $N$ .

### 2.3 Predictive Variance Decomposition and Out-of-Distribution Calibration

An important consideration in sparse variational Gaussian process inference is the proper treatment of predictive variance, particularly for out-of-distribution (OOD) detection. The exact predictive variance at a test point  $x_*$  can be decomposed into two geometrically interpretable components [15,34]:

$$\text{Var}(f_*) = \underbrace{\mathbf{k}_{x_*}^\top \mathbf{K}_{\mathbf{ZZ}}^{-1} \text{Cov}(\mathbf{u}) \mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{k}_{x_*}}_{\text{Projected Variance (V}_{\text{proj}})} + \underbrace{[k(x_*, x_*) - \mathbf{k}_{x_*}^\top \mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{k}_{x_*}]}_{\text{Orthogonal Variance (V}_{\text{orth}})}$$

The projected variance  $V_{\text{proj}}$  quantifies uncertainty transmitted through the inducing point subspace. That is, it captures how posterior uncertainty over  $\mathbf{u}$  propagates to predictions at  $x_*$ . The orthogonal variance  $V_{\text{orth}}$ , identified as the Nyström approximation error, measures information lost by the sparse representation: the component of  $f_*$  that lies outside the span of the inducing functions.

Standard implementations of moment-matching in deep GP architectures often retain only  $V_{\text{proj}}$ , implicitly assuming the inducing points provide sufficient coverage of the input domain. This omission leads to variance collapse for out-of-distribution inputs: as a test point  $x_*$  moves far from all inducing locations  $\mathbf{Z}$ , the cross-covariance  $k(x_*, \mathbf{Z}) \rightarrow \mathbf{0}$ , causing projected variance to vanish regardless of true epistemic uncertainty. We address this by incorporating the Nyström approximation error  $V_{\text{orth}} = k(x_*, x_*) - k(x_*, \mathbf{Z})K_{\mathbf{ZZ}}^{-1}k(\mathbf{Z}, x_*)$  into our predictive variance, approximated efficiently via the  $\psi$ -statistics as  $V_{\text{orth}} \approx \sigma_f^2(1 - \sum_m \psi_1^2 / \sigma_f^4)$ . This guarantees that as  $x_* \rightarrow \infty$ , predictive variance recovers the prior  $\sigma_f^2$ , providing a geometric safeguard against silent extrapolation failures essential for the out-of-distribution detection as examined in Study C.

A property of this decomposition with practical significance is that the orthogonal variance operates independently of the variational posterior parameters and, as a result, independently of the KL divergence weight  $\lambda$  in the ELBO objective. The projected variance  $V_{\text{proj}}$  depends on  $\text{Cov}(\mathbf{u})$ , which is shaped by the balance between reconstruction fidelity and prior regularization during training. However, the orthogonal variance  $V_{\text{orth}} = k(x_*, x_*) - \mathbf{k}_{x_*}^\top \mathbf{K}_{\mathbf{ZZ}}^{-1} \mathbf{k}_{x_*}$  depends only on the kernel function and the inducing point locations, not on the variational distribution  $q(\mathbf{u})$ . With this decoupling, the KL weight can be selected based on calibration requirements (e.g., for the heteroscedastic noise estimation in Study A) without compromising the geometric OOD detection guarantee that  $V_{\text{orth}} \rightarrow \sigma_f^2$  as  $x_*$  departs from the inducing support. Indeed, moderate KL regularization may improve OOD detection reliability by anchoring the variational posterior to the GP prior, stabilizing the projected variance component and preventing degenerate solutions that can arise in the unregularized ( $\lambda = 0$ ) setting (Section 4.3, Table 2).

## 2.4 Analytic Moment Matching for Uncertainty Propagation

A fundamental challenge in deep Gaussian process architectures concerns the propagation of uncertainty through layers. When the input to a Gaussian process is itself a random variable with distribution  $p(x)$ , evaluating the posterior predictive distribution requires computing expectations over this input distribution. For arbitrary input distributions and kernels, these expectations are generally intractable. However, the univariate structure of Kolmogorov-Arnold edges and the additive aggregation operation enable tractable approximations through analytic moment matching.

Our approach relies on a mean-field approximation: we assume statistical independence between the edge function outputs within each layer. That is, for a layer with

input  $\mathbf{x} \in \mathbb{R}^{D_{\text{in}}}$  and edge functions  $\{\phi_{j,i}\}$ , we approximate the joint output distribution by the product of marginals. This assumption is natural in the Kolmogorov-Arnold topology because each edge function  $\phi_{j,i}(x_i)$  operates on a single input dimension, and the variational posteriors over inducing variables are specified independently for each edge. The mean-field factorization enables the additive variance decomposition presented below and is exact when the GP priors across edges are independent. We note, however, that the validity and limitations of the mean-field approximation, in particular, as applied to SVGP-KAN, have yet to be rigorously established.

One primary limitation arises at inter-layer boundaries: when the output of one layer serves as input to the next, the true joint distribution over layer outputs may exhibit correlations induced by shared inputs, which the mean-field approximation neglects. The mean field approximation has been successfully implemented in deep GP architectures [25], when layers have moderate width and the additive structure limits the degree of inter-output dependence.

Consider a Gaussian process  $\phi$  with squared exponential kernel parameterized by signal variance  $\sigma_f^2$  and lengthscale  $\ell$ . Suppose the input  $x$  follows a Gaussian distribution  $x \sim \mathcal{N}(\mu_x, \sigma_x^2)$ . We seek to compute the mean and variance of the output distribution  $p(y) = \int p(y | x)p(x)dx$  where  $p(y | x)$  represents the posterior predictive distribution of the Gaussian process.

Following [29], the expected posterior mean can be computed as:

$$\mathbb{E}_{p(x)}[\mu_\phi(x)] = \mathbf{m}^\top \mathbf{K}_{ZZ}^{-1} \Psi_1$$

where the vector  $\Psi_1 \in \mathbb{R}^M$  has components:

$$[\Psi_1]_m = \mathbb{E}_{p(x)}[k(x, z_m)] = \sigma_f^2 \left( \frac{\ell^2}{\ell^2 + \sigma_x^2} \right)^{1/2} \exp \left( -\frac{(z_m - \mu_x)^2}{2(\ell^2 + \sigma_x^2)} \right)$$

This expression reveals a regularization property inherent in uncertainty propagation. As input uncertainty  $\sigma_x^2$  increases, the effective lengthscale becomes  $\sqrt{\ell^2 + \sigma_x^2}$ , resulting in smoother effective functions. This automatic smoothing prevents overconfident extrapolation by increasing prediction uncertainty in regions where inputs are uncertain.

The variance of the output distribution comprises two terms reflecting distinct sources of uncertainty:

$$\mathbb{V}_{p(x)}[y] = \mathbb{E}_{p(x)}[\sigma_\phi^2(x)] + \mathbb{V}_{p(x)}[\mu_\phi(x)]$$

The first term captures epistemic uncertainty arising from finite training data, while the second reflects uncertainty propagation from stochastic inputs. For the additive structure of Kolmogorov-Arnold layers where  $y = \sum_{i=1}^{D_{\text{in}}} \phi_i(x_i)$ , the total output variance under a mean-field (layer-wise independence) approximation becomes:

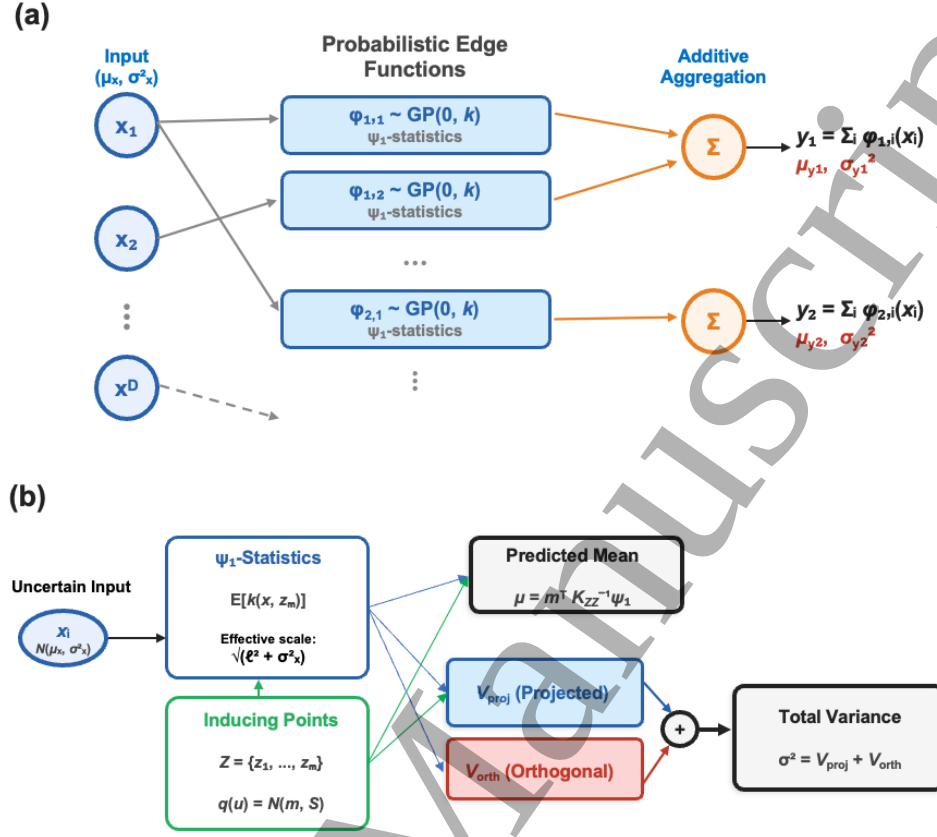
$$\mathbb{V}[y] = \sum_{i=1}^{D_{\text{in}}} (\mathbb{E}[\sigma_{\phi_i}^2(x_i)] + \mathbb{V}[\mu_{\phi_i}(x_i)])$$

This additive decomposition enables efficient propagation through deep networks while maintaining interpretability, as uncertainty contributions from individual input dimensions can be isolated and analyzed.

As discussed in Section 2.3, the epistemic uncertainty term  $\mathbb{E}[\sigma_{\phi}^2(x)]$  incorporates both projected and orthogonal variance components, ensuring that predictive uncertainty correctly increases for inputs far from the inducing point support.

Figure 1 provides a schematic overview of the architecture of SVGP-KAN. Figure 1(a) illustrates the layer-level structure. Each input dimension  $x_i$  connects to each output dimension  $y_j$  through an independent univariate Gaussian process edge function  $\phi_{j,i}(x_i)$ , replacing the scalar weights of a standard linear layer. Outputs are obtained by additive aggregation:  $y_j = \sum_i \phi_{j,i}(x_i)$ . Each edge function carries its own kernel hyperparameters (lengthscale, signal variance) and variational parameters (inducing points, posterior mean and covariance), enabling per-connection uncertainty quantification and automatic relevance determination.

Figure 1(b) details the moment-matching computation for a single edge, showing how the  $\psi_1$ -statistics mediate between the input distribution and the predictive moments. Given an input distribution  $\mathcal{N}(\mu_x, \sigma_x^2)$ , the  $\psi_1$ -statistics (blue) integrate the kernel function against the input distribution to produce the predicted mean and variance decomposition. The total predictive variance separates into a projected component  $V_{\text{proj}}$ , which depends on the variational posterior  $q(\mathbf{u})$  (green arrows), and an orthogonal component  $V_{\text{orth}}$ , which depends only on the kernel function and inducing point geometry. This decoupling — visible in the arrow structure as the absence of a green pathway to  $V_{\text{orth}}$  — ensures that out-of-distribution detection through  $V_{\text{orth}}$  is independent of the KL regularization weight  $\lambda$ .



**Figure 1: Schematic of a Single GPKAN Layer.** (a) Layer-level structure. (b) Moment-matching computation for a single edge.

## 2.5 Observation Noise and Aleatoric Uncertainty

In many scientific applications, measurements are corrupted by noise whose intensity varies spatially or temporally. Heteroscedastic observation noise, where variance depends on input location or system state, requires explicit modeling to avoid biased predictions and miscalibrated uncertainties. We extend the framework to incorporate input-dependent observation noise through a secondary Gaussian process.

Let  $f(\mathbf{x})$  denote the latent function of interest and  $\sigma_{\text{noise}}^2(\mathbf{x})$  represent the observation noise variance at location  $\mathbf{x}$ . The likelihood becomes:

$$p(y | \mathbf{x}, f, \sigma_{\text{noise}}^2) = \mathcal{N}(y; f(\mathbf{x}), \sigma_{\text{noise}}^2(\mathbf{x}))$$

To ensure positivity, we model the log-variance through a separate Gaussian process:

$$\log \sigma_{\text{noise}}^2(\mathbf{x}) \sim \mathcal{GP}(m_{\text{noise}}(\mathbf{x}), k_{\text{noise}}(\mathbf{x}, \mathbf{x}'))$$

where  $m_{\text{noise}}$  represents a mean function (often set to a learnable constant) and  $k_{\text{noise}}$  denotes a kernel function that may differ from the kernel used for the latent function.

During training, both the latent function and noise variance are inferred jointly through the variational framework. The expected log-likelihood term in the ELBO becomes:

$$\mathbb{E}_{q(f)q(\log\sigma^2)} \left[ \sum_{n=1}^N \log \mathcal{N}(y_n; f(\mathbf{x}_n), \sigma^2(\mathbf{x}_n)) \right]$$

This expectation can be approximated using sampling or quadrature. The resulting framework naturally separates aleatoric uncertainty (inherent stochasticity in observations) from epistemic uncertainty (model uncertainty due to finite data), which in principle can provide calibrated confidence intervals that correctly expand in low-data regions and contract where observations are plentiful and precise.

## 2.6 Computational Complexity and Scalability

The computational advantages of Sparse Variational Gaussian Process Kolmogorov-Arnold Networks (SVGP KANs) arise from the factorization of edge functions and inducing point approximations. Consider a network with  $L$  layers, where layer  $l$  maps dimension  $D_l$  to  $D_{l+1}$ . The total number of edge functions across the network is  $\sum_{l=1}^L D_l D_{l+1}$ . For each edge function, maintaining variational parameters requires storing  $M$  inducing inputs, an  $M$ -dimensional mean vector, and an  $M \times M$  diagonal covariance matrix, totaling  $O(3M)$  parameters per edge.

Forward propagation through a single layer requires computing posterior means and variances for  $D_{l+1}$  outputs, each involving  $D_l$  edge evaluations. Each edge evaluation requires matrix-vector products with  $O(M^2)$  complexity, yielding  $O(D_l D_{l+1} M^2)$  per-layer cost. For typical architectures where  $M$  ranges from 20 to 100, this remains tractable even for networks with thousands of parameters. Importantly, the cost scales linearly with training set size  $N$  when using mini-batch stochastic optimization, contrasting with  $O(N^3)$  scaling in exact Gaussian process inference.

The inducing inputs  $\mathbf{Z}_{j,i}$  may be initialized uniformly across the input domain or through k-means clustering of training data. During optimization, these inducing inputs can be treated as learnable parameters, enabling the model to automatically position pseudo-datapoints in regions of high variability or importance. Gradient-based optimization of inducing locations has been shown to improve approximation quality in sparse Gaussian process literature. But it also has the potential to negatively impact the validity of the mean field approximation.

### 3. Experimental Design and Implementation

#### 3.1 Architecture Specifications

We implement the sparse variational Gaussian process Kolmogorov-Arnold network framework in PyTorch. The core architectural component, termed GPKANLayer, replaces standard linear transformations with sparse variational Gaussian process mappings. Each layer maintains inducing inputs stored as learnable parameters of dimension  $(D_{\text{in}} \times D_{\text{out}}) \times M$ , variational means of the same dimension, and diagonal variational variances stored in the log-space. The per-layer KL divergence takes on the mean value over all  $D_{\text{in}} \times D_{\text{out}}$  edges, ensuring that the KL weight  $\lambda$  need not be re-tuned when changing layer widths.

For applications requiring convolutional architectures, we construct encoder-decoder networks where dense Kolmogorov-Arnold layers form bottlenecks between convolutional feature extractors and generators. The encoder processes input fields through convolutional layers with kernel sizes adapted to the spatial scale of features, followed by flattening and projection through a GPKANLayer bottleneck. The decoder performs the inverse operation, mapping latent representations back to the spatial domain through transposed convolutions.

Kernel functions are implemented with learnable hyperparameters including length scales and signal variances. We employ log-parameterization for positive hyperparameters to ensure valid parameter spaces during unconstrained optimization. Initial length scale values are set based on the typical range of input features, with signal variances initialized to unity. For heteroscedastic noise modeling, we introduce separate Gaussian process priors over log-noise-variance, enabling the model to learn spatially varying observation uncertainty from data.

All experiments reported in the present paper were implemented in Python 3.13 on a MacBook Air with Apple M2 CPU (8-core, 8 GB unified memory); no GPU acceleration was used. Random seeding was applied as follows. For Study A, the training dataset and test set were generated with fixed NumPy and PyTorch random number seeds of 42 for training data and 999 for the held-out test set, ensuring identical data exposure across all baseline methods compared in Table 2. Each model (SVGP-KAN, both MLP variants, MC Dropout, and all five Deep Ensemble members) was initialized with a PyTorch seed of 42 immediately before training. For Study B, the ground truth trajectory and the physics sensitivity test were generated with a NumPy random number seed of 42; PyTorch model weight initialization is not separately seeded, so minor run-to-run variation from default initialization is possible. For Study C (Tables 3 and 4), all models were initialized with a PyTorch random number seed of 7. For the KL weight ablation (Table 2), five independent training runs per  $\lambda$  setting were executed with random number seeds of 7, 49, 91, 133, and 175 for runs 0–4, respectively; the standard deviations in Table 2 reflect variance across these five runs. MC Dropout uncertainty was estimated from 20 stochastic forward passes in Study A and 30 passes in Study C; Deep Ensembles comprised 5 independently

initialized members in both studies. Computational cost measurements in Table 4 are averaged over 50 training epochs, and 100 inference passes after a 5-epoch warmup period.

### 3.2 Study A: Calibration of Heteroscedastic Measurement Uncertainty

The first study evaluates the framework’s ability to identify and calibrate spatially varying observation noise in single-step prediction of a two-dimensional scalar field. We construct a synthetic dataset in which measurement noise scales with the local field amplitude, creating a controlled test case for heteroscedastic uncertainty estimation where the ground truth noise structure is known exactly.

The scalar field  $\omega(x, y)$  is defined on a periodic domain  $[0, 2\pi]^2$  and initialized as a smooth single-mode pattern perturbed by small random fluctuations:

$$\omega(x, y) = \sin(x)\cos(y) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.01)$$

The dynamics consist of uniform diagonal advection (a one-grid-cell shift per timestep in each spatial direction) followed by linear damping with coefficient  $\alpha = 0.95$ . Observations of the evolved field are corrupted by heteroscedastic Gaussian noise whose standard deviation scales with the local field amplitude:

$$\omega_{n+1} = \alpha \mathcal{A}[\omega_n] + \sigma_{\text{obs}}(\omega) \eta, \quad \eta \sim \mathcal{N}(0, 1)$$

where  $\mathcal{A}$  denotes the diagonal advection operator and the observation noise level is

$$\sigma_{\text{obs}}(\omega) = 0.01 + 0.15 |\omega|$$

This amplitude-dependent noise structure simulates measurement scenarios common in experimental fluid mechanics, where sensor precision degrades in regions of strong flow activity — for example, particle image velocimetry measurements in high-shear zones or hot-wire anemometry in highly turbulent regions. Regions where  $|\omega|$  is large carry proportionally higher measurement noise, while quiescent regions (near zero crossings of the field) have near-constant low noise. The model must learn this amplitude-dependent noise structure from data alone, without access to the noise-generating function.

The network architecture is a convolutional encoder-decoder with a single input channel, 16 base feature channels, and a GPKAN Layer bottleneck with 32 inducing points per edge. The encoder consists of two convolutional blocks with  $3 \times 3$  kernels and circular padding (respecting the periodic domain), each followed by  $2 \times 2$  max pooling. The decoder mirrors this structure using transposed convolutions with skip connections from the encoder. The final convolutional layer outputs two channels: a predictive mean  $\hat{\mu}(x, y)$  and a learned log-variance  $\log \hat{\sigma}^2(x, y)$ , enabling the model to express spatially varying predictive uncertainty. Training data is generated on-the-fly at each epoch on a  $64 \times 64$  spatial grid with batch size 16, so the model never sees the same noise realization twice. The loss function is the Gaussian negative log-likelihood under the learned

heteroscedastic model, augmented with KL divergence regularization weighted by  $\lambda = 0.01$ . Evaluation is performed on 100 independent test samples, totaling 409,600 prediction points.

All baseline methods we used for comparison share the identical convolutional encoder-decoder backbone as SVGP-KAN and are trained with the same Adam optimizer, learning rate of  $2 \times 10^{-3}$ , training duration of 300 epochs (70 epochs for Study C), and batch size of 16, ensuring an equal training budget. For Monte Carlo dropout, a dropout rate of  $p = 0.15$  is applied after each encoder convolutional block and within the bottleneck layer in Study A (for Study C,  $p = 0.20$ ). For deep ensembles,  $M = 5$  independently initialized members are trained; we compute the predictive mean and variance by mixing the per-member Gaussian distributions. The architecture-matched MLP bottleneck uses a two-layer MLP with hidden dimension matching the SVGP-KAN layer structure; the parameter-matched MLP uses a wider hidden dimension to match SVGP-KAN's total parameter count (27,410 for Study A; 82,017 for Study C). All baseline hyperparameters were set to standard values from the respective reference implementations; no additional hyperparameter optimization was performed beyond the learning rate, which was validated on a held-out 10% subset of the training data using the same protocol for all methods.

### 3.3 Study B: Epistemic Uncertainty in Multi-Step Forecasting

The second study examines epistemic uncertainty growth in sequential predictions of advection-diffusion transport phenomena. Unlike the first study where uncertainty primarily reflects aleatoric measurement noise, this investigation focuses on model uncertainty arising from approximation errors in the neural network that compound over multiple prediction steps, even when the underlying physical dynamics remain stable.

We simulate two-dimensional advection-diffusion dynamics governing the evolution of a scalar temperature field  $T(\mathbf{x}, t)$  under the partial differential equation:

$$\frac{\partial T}{\partial t} = -\mathbf{u} \cdot \nabla T + \kappa \nabla^2 T + S(\mathbf{x}, t)$$

where  $\mathbf{u}(\mathbf{x}, t)$  denotes a prescribed velocity field,  $\kappa$  represents the thermal diffusivity, and  $S$  incorporates source and sink terms. The velocity field is assumed to evolve slowly in time according to a time-dependent stream function:

$$\psi(\mathbf{x}, t) = \sin(2x + \omega t)\cos(y) + 0.5\cos(x)\sin(2y - 0.5\omega t)$$

where  $\omega = 0.1$  controls the temporal evolution rate. This creates a gradually varying flow pattern where different initial conditions experience different advective transport histories.

We again use a 64x64 spatial grid with periodic boundary conditions. Forward Euler time integration with timestep  $\Delta t = 0.1$  advances the system for 15 steps, providing ground truth trajectories. To create an advection-dominated regime that tests the model's predictive capabilities, we set the advection velocity magnitude to 0.5 and the diffusivity to

$\kappa = 0.002$ , yielding a Péclet number  $Pe \approx 1571$ . The source term amplitude is 0.02. The model is trained solely on single-step transitions,  $(T_n \rightarrow T_{n+1})$ , reflecting scenarios where only short-term measurements are available but long-term forecasts are required.

The physical system’s sensitivity to initial conditions was tested independently by evolving two trajectories with 0.1% initial perturbation using the ground truth PDE solver. The mean difference at  $t = 14$  remained essentially constant at the initial difference, confirming that the system is stable rather than chaotic. This stability analysis is critical for interpreting subsequent results: any observed uncertainty growth in model predictions reflects epistemic limitations of the neural network approximation, not sensitive dependence on initial conditions in the underlying physics.

The SVGP-KAN architecture employs a convolutional encoder-decoder structure. The encoder uses two convolutional layers with ReLU activations, reducing spatial dimensions. A GPKANLayer bottleneck with 24 base channels and 32 inducing points per edge compresses the representation. The decoder mirrors the encoder structure using transposed convolutions to reconstruct the temperature field. Training proceeds over 1000 epochs using Gaussian negative log-likelihood loss augmented with KL divergence regularization weighted by  $\lambda = 0.01$ , employing the Adam optimizer with learning rate  $2 \times 10^{-3}$ .

Uncertainty quantification in multi-step forecasting is performed through ensemble rollout. We generate an ensemble of  $n_{\text{ens}} = 10$  initial conditions by perturbing a reference initial state with 1% Gaussian noise. Each ensemble member is propagated forward for 15 steps using the posterior mean prediction (deterministic propagation) at each timestep, with predictions clipped to the physical domain  $[-2, 2]$  to prevent numerical instabilities. The ensemble spread, measured through pixelwise standard deviation across ensemble members, quantifies prediction uncertainty at each temporal horizon.

### 3.4 Study C: Out-of-Distribution Detection in Convolutional Autoencoders

The third study evaluates anomaly detection capabilities by integrating SVGP-KANs as bottleneck layers in convolutional autoencoders. The task requires distinguishing in-distribution inputs that the model was trained on from anomalous out-of-distribution inputs that should trigger high uncertainty alerts.

This example study can be extended and applied to a wide variety of applications in science and technology, including bio imaging. As a notional demonstration, we use the readily accessible MNIST dataset of handwritten digits, training the autoencoder exclusively on images of the digit “0”. The architecture follows a standard encoder-decoder design: the encoder applies two convolutional layers (8 filters, then 16 filters, each with 3x3 kernels and stride 2) followed by ReLU activations, reducing spatial dimensions from  $28 \times 28$  to  $7 \times 7$ . The flattened feature map of dimension  $16 \times 7 \times 7 = 784$  is projected through a linear layer to 64 dimensions, then through a GPKANLayer bottleneck to a 6-dimensional latent code. The decoder mirrors the encoder, first expanding the latent code to 784 dimensions through a linear layer, reshaping to spatial dimensions, then

applying transposed convolutions to reconstruct 28x28 images. The GPKANLayer bottleneck employs 20 inducing points per edge with RBF kernels. Length scale and signal variance hyperparameters are initialized to moderate values and remain trainable, allowing the model to learn appropriate uncertainty scales from data.

Training proceeds over 70 epochs using mean squared error reconstruction loss augmented with KL divergence regularization weighted by  $\lambda = 0.001$ . The orthogonal variance correction described in Section 2.3 ensures that predictive variance correctly reverts to the prior when the encoder projects out-of-distribution features into regions far from inducing points. This enables principled Bayesian inference while maintaining out-of-distribution detection capabilities.

At test time, we present both normal “0” digit images and anomalous “7” digit images. For each input, we compute the reconstruction error and the bottleneck uncertainty, defined as the sum of predicted variances across the 6 latent dimensions. The hypothesis is that anomalous inputs will yield both higher reconstruction errors (the decoder cannot accurately generate “7” patterns from latent codes trained on “0” patterns) and higher bottleneck uncertainties. The encoder encounters feature combinations unseen during training, triggering high epistemic uncertainty in the Gaussian process edges due to reversion to the prior variance through the orthogonal variance mechanism.

Quantitative evaluation measures the anomaly score ratio, that is, the uncertainty on anomalous inputs divided by the uncertainty on normal inputs. A well-calibrated model should produce ratios significantly greater than unity, indicating clear discrimination. We further evaluate discrimination quality through receiver operating characteristic (ROC) analysis, computing the area under the curve (AUC) to assess overall separation quality. We also examine reconstruction quality visually, testing whether anomalous inputs produce characteristic degraded or smoothed reconstructions reflecting the model’s attempt to project out-of-distribution features onto the learned manifold of normal data.

### 3.5 Computational Overhead

The computational overhead introduced by sparse variational inference is expected to remain modest relative to deterministic implementations of the KAN architecture. For edge functions with  $M = 20$  inducing points, the variational parameters (mean vector and diagonal covariance matrix) add approximately  $3M = 60$  parameters per edge. A deterministic B-spline implementation with grid size  $G = 20$  and order  $K = 3$  stores  $(G + K)$  coefficients per edge, totaling 23 parameters. Thus, the probabilistic formulation increases parameter count by roughly 3-fold per edge. However, this comparison understates the computational advantage: deterministic B-splines require recursive evaluation during forward passes, while inducing point formulations employ simple matrix-vector products amenable in principle to GPU parallelization.

## 4. Results

### 4.1 Study A: Heteroscedastic Noise Calibration in Scalar Field Reconstruction

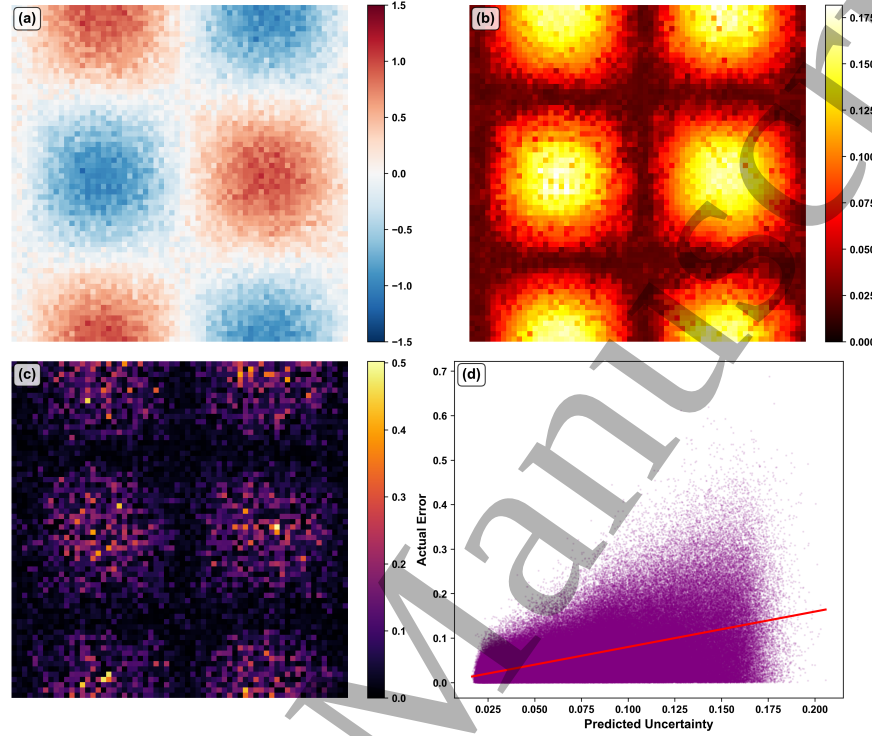
The model’s ability to infer spatially varying noise patterns was evaluated on the scalar field prediction task with amplitude-dependent heteroscedastic measurement noise to demonstrate that KL-regularized training has the ability to produce well-calibrated uncertainty estimates. The Pearson correlation between predicted uncertainty and actual absolute error was  $\rho = 0.45$ , indicating statistically significant learning of heteroscedastic patterns. The moderate correlation reflects the stochastic nature of the problem, that is, the predicted uncertainty represents expected error magnitude, not deterministic error values. The correlation is strong enough to demonstrate that the model has successfully learned to identify high-noise regions, yet not perfect as would be inappropriate for a system with inherent measurement noise.

Coverage statistics demonstrated good calibration across multiple confidence levels, below 1 percentage point at  $\pm 1\sigma$ ,  $\pm 2\sigma$ , and  $\pm 3\sigma$ . The calibration error at  $2\sigma$  was 0.1 percentage points, well within acceptable bounds for practical uncertainty quantification. Standardized prediction errors  $(\hat{\omega} - \mu)/\sigma$  exhibited mean of -0.09 and standard deviation 1.01, closely approximating the ideal  $\mathcal{N}(0,1)$  distribution expected for well-calibrated probabilistic predictions. The near-zero mean confirms unbiased predictions. Overall RMSE of 0.085 suggests accurate mean predictions alongside well-calibrated uncertainty estimates.

To quantitatively assess the model’s ability to distinguish aleatoric from epistemic uncertainty, we compare the predicted uncertainty field against the known ground truth noise level. Since the synthetic data is generated with heteroscedastic noise  $\sigma_{\text{obs}} = 0.01 + 0.15|\omega|$ , the ground truth noise standard deviation is available at each spatial location. We compute the Pearson correlation between the predicted standard deviation field  $\hat{\sigma}(x, y)$  and the absolute prediction error  $|\omega_{\text{true}} - \hat{\omega}|$  across all spatial locations and test samples. This yields  $\rho = 0.45$ , confirming that the model has learned to assign higher uncertainty to regions where errors are larger. We note that the predicted uncertainty captures aleatoric contributions (measurement noise magnitude) rather than simply producing uniform error estimates. In regions where the field amplitude  $|\omega|$  is large, both the true noise level and the predicted uncertainty increase proportionally, suggesting that the heteroscedastic output head has successfully learned the amplitude-dependent noise structure. This can be contrasted with the epistemic uncertainty behavior demonstrated in Study B, where uncertainty grows temporally due to model approximation error rather than measurement noise, providing complementary evidence that the framework can in theory correctly decompose total uncertainty into its constituent sources.

Figure 2 visualizes the results from the first study. The predicted uncertainty field exhibits spatial structure corresponding to the ground truth heteroscedastic noise pattern, with elevated uncertainty in the designated high-noise regions. The calibration scatter plot shows a strong linear relationship between predicted uncertainty and actual error. The

deviation from perfect linearity reflects appropriate acknowledgment of irreducible aleatoric noise rather than miscalibration.



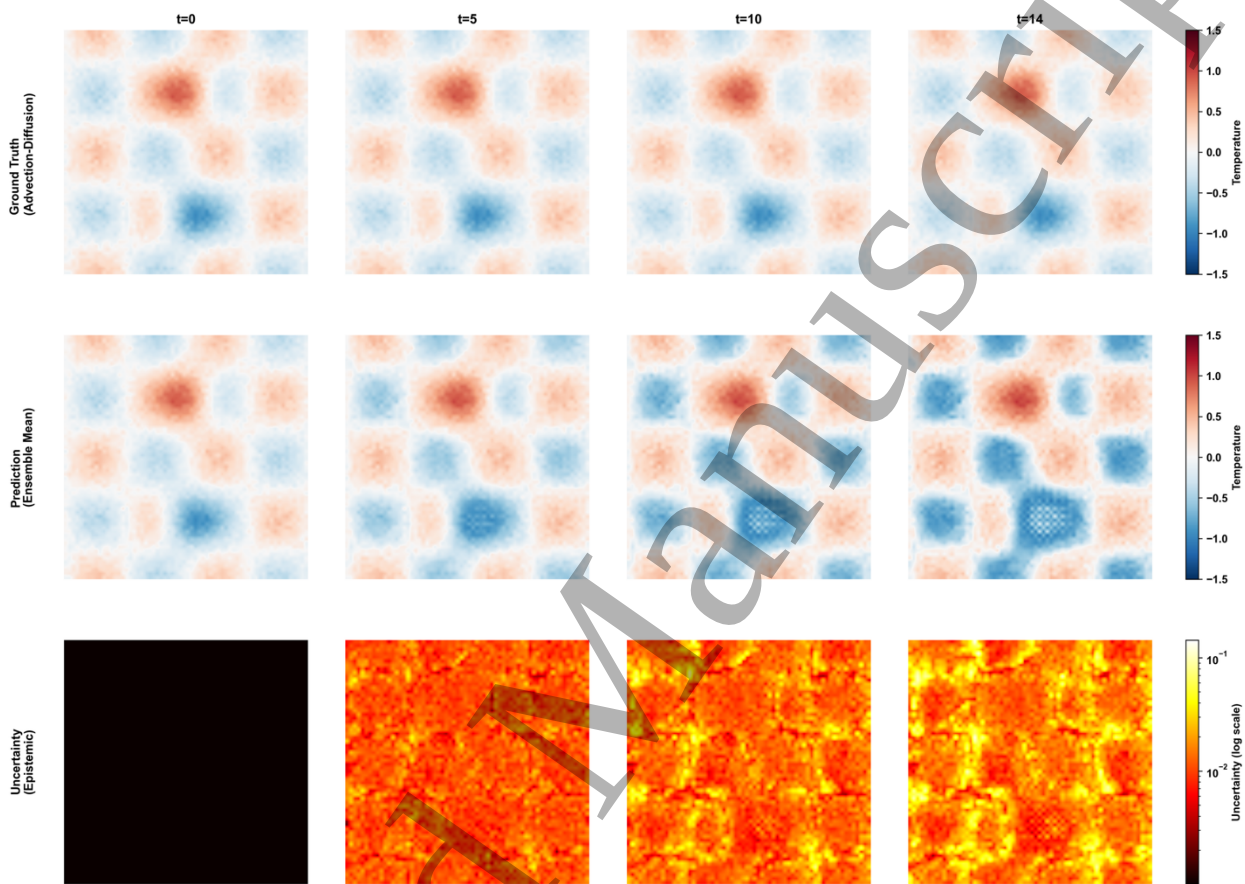
**Figure 2: Heteroscedastic Noise Calibration in Scalar Field Prediction.** Spatially varying uncertainty calibration for the single-step prediction task. (a) Predicted scalar field  $\hat{\omega}(x, y)$  showing characteristic high (red) and low (blue) regions inherited from the  $\sin(x)\cos(y)$  base pattern. (b) Predicted uncertainty  $\hat{\sigma}(x, y)$  learned by the heteroscedastic output head; elevated uncertainty corresponds to regions of large field amplitude, consistent with the ground truth noise model  $\sigma_{\text{obs}} = 0.01 + 0.15|\omega|$ . (c) Absolute prediction errors  $|\omega_{\text{true}} - \hat{\omega}|$ . (d) Calibration scatter plot: predicted standard deviation versus absolute error across all spatial locations, with Pearson correlation  $\rho = 0.45$ .

## 4.2 Study B: Multi-Step Prediction Uncertainty Growth in Advection-Diffusion

The time-dependent advection-diffusion system reveals characteristic epistemic uncertainty growth patterns in multi-step forecasting of stable dynamics. Figure 3 displays the evolution over a 15-step forecast horizon, showing ground truth temperature fields, ensemble mean predictions, and ensemble spread (epistemic uncertainty).

At the initial timestep  $t = 0$ , the ground truth exhibits localized hot and cold regions arranged in a regular pattern. The ensemble mean prediction closely matches this initial condition, as expected given the small 1% initial perturbations. The ensemble spread at

$t = 0$  remains nearly uniform at low amplitude (black in visualization), reflecting only the initial perturbation magnitude.



**Figure 3: Multi-Step Prediction Uncertainty Growth in Advection-Diffusion Dynamics:** Three-row visualization tracking epistemic uncertainty evolution over 15-step forecast horizon on 64x64 grid. Top row (Ground Truth), Center row (Prediction), and Bottom row (Uncertainty - Epistemic). Note: Uncertainty colormap uses logarithmic scaling to ensure visibility of early-timestep uncertainty structure.

As the forecast progresses to  $t = 5$ , the ground truth concentration field evolves under combined diagonal advection and diffusion, with initial patterns transported and smoothed. The ensemble mean prediction tracks this evolution, though subtle differences begin to emerge. The ensemble spread increases, with localization beginning to appear in regions of steep gradients where small errors in predicted advection velocities compound.

At  $t = 10$ , divergence between predictions and ground truth becomes more pronounced. While large-scale structure remains qualitatively similar, quantitative differences in temperature magnitudes and spatial positions accumulate. The ensemble

spread grows, exhibiting spatial heterogeneity with highest uncertainty concentrated along interfaces between hot and cold regions.

By the final forecast step  $t = 14$ , the ensemble mean prediction retains correlation with ground truth but exhibits noticeable smoothing and phase offsets. The uncertainty quantification accurately captures this degradation: ensemble spread reaches peak values with strong spatial organization indicating where model predictions have become unreliable (higher uncertainty).

To complement the qualitative uncertainty visualization with a standard accuracy metric, Table 1 reports the root-mean-square error (RMSE) between the model's mean prediction and the ground truth field at each forecast step, alongside the mean GP bottleneck variance. The GP variance is the SVGP-KAN bottleneck's own posterior variance, that is, the direct output of the sparse variational GP layer at each autoregressive step, rather than inter-trajectory spread from perturbed initial conditions, making it an intrinsic property of the model's probabilistic state. Both RMSE and mean GP variance increase monotonically over the 15-step horizon (Pearson  $\rho = 0.999$ ), confirming that the GP uncertainty signal correctly tracks the direction of prediction error accumulation. RMSE grows from 0.007 at  $t = 1$  to 0.131 at  $t = 15$ , while mean GP variance grows from 4.20 to 4.98 (1.19-fold). The modest absolute growth in GP variance reflects that it is expressed in the prior signal scale  $\sigma_r^2$  rather than in prediction-error units; what is significant is the consistent monotonic increase as the autoregressive rollout progressively moves bottleneck inputs away from the training distribution, causing the GP posterior to revert toward the prior.

The 12.7-fold spatial variation in ensemble spread at  $t = 14$  (Figure 3) shows that the uncertainty concentrates in spatially structured regions rather than being uniform across the domain, consistent with the theoretical expectation that uncertainty grows where model inputs deviate most from training support, specifically, at advective flow interfaces where steep gradients challenge the model's interpolation. The physics sensitivity test confirmed that the underlying dynamics are stable (1.0x sensitivity ratio), so the observed uncertainty growth reflects model-induced epistemic limitations rather than physical chaos.

**Table 1:** Study B Per-Step Prediction Accuracy and GP Epistemic Uncertainty

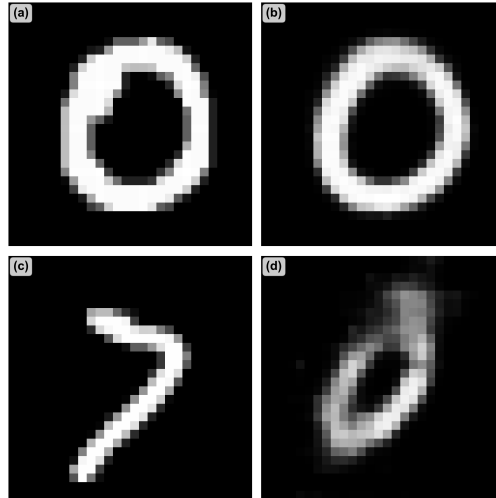
| Timestep, $t$ | RMSE  | Mean GP Var |
|---------------|-------|-------------|
| 1             | 0.007 | 4.197       |
| 5             | 0.038 | 4.367       |
| 10            | 0.079 | 4.648       |
| 15            | 0.131 | 4.983       |

### 4.3 Study C: Anomaly Detection through Bottleneck Uncertainty

The convolutional autoencoder with SVGP-KAN bottleneck has out-of-distribution detection capabilities. Figure 4 shows visual comparison of normal and anomalous reconstructions.

The detection mechanism arises from the predictive variance decomposition described in Section 2.3. When presented with a “0” digit from the training distribution, the autoencoder reconstructs the input with high fidelity (top row, Fig. 4). The bottleneck uncertainty, measured as the sum of predicted variances across six latent dimensions, remains low for in-distribution samples. The variational posterior for inducing variables concentrates in regions of the GP input space corresponding to “0” features, and the projected variance  $V_{\text{proj}}$  dominates the total predictive variance since inputs lie close to well-covered regions of the inducing point support.

In contrast, when processing a “7” digit unseen during training (bottom row, Fig. 4), reconstruction quality degrades substantially as the decoder attempts to force anomalous features onto the learned manifold of “0” patterns. Bottleneck uncertainty increases substantially for out-of-distribution inputs because the encoder projects “7” features into regions far from any inducing points. In these regions, the cross-covariance  $\mathbf{k}_{x_*}$  vanishes, causing the projected variance to approach zero. However, the orthogonal variance term  $V_{\text{orth}} \rightarrow \sigma_f^2$  ensures that the total predictive variance reverts to the prior, providing the elevated uncertainty signal that enables anomaly detection. This geometric mechanism, guaranteed in theory by the variance decomposition rather than dependent on specific hyperparameter choices, may eliminate the need for explicit anomaly detection training while providing principled uncertainty estimates that identify unfamiliar inputs.



**Figure 4: Out-of-Distribution Detection via Bottleneck Uncertainty (Study C):**

Comparison demonstrating anomaly detection through the orthogonal variance mechanism in SVGP-KAN bottleneck. Top row (Normal Input - Digit “0”), Bottom row (Anomalous Input - Digit “7”). Left: Input image. Right: Reconstruction.

To assess sensitivity to the KL regularization weight, we conducted an ablation study varying  $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$  with five independent training runs per setting. Table 2 reports the mean and standard deviation of ROC-AUC, uncertainty ratio, and separation percentage for the Study C OOD detection task.

The results reveal that moderate KL regularization improves OOD detection reliability rather than hindering it. Without regularization ( $\lambda = 0$ ), detection performance degrades to chance level ( $\text{ROC-AUC} = 0.50 \pm 0.22$ ), as the variational posterior drifts freely from the GP prior, destabilizing the projected variance component  $V_{\text{proj}}$  and leading to degenerate latent representations. Introducing even a small regularization weight ( $\lambda = 10^{-4}$ ) substantially improves detection ( $\text{ROC-AUC} = 0.82 \pm 0.09$ ), with the best performance at  $\lambda = 10^{-3}$  ( $\text{ROC-AUC} = 0.92 \pm 0.03$  and the tightest variance across runs). Performance remains robust across the entire non-zero range tested, with ROC-AUC above 0.82 for all  $\lambda \geq 10^{-4}$ . These findings are consistent with the theoretical analysis in Section 2.3: the orthogonal variance  $V_{\text{orth}}$ , which provides the primary OOD detection signal, depends only on the kernel and inducing point geometry — not on the variational posterior or KL weight. The role of KL regularization is to stabilize  $V_{\text{proj}}$  by anchoring the posterior to the GP prior, preventing the pathological behavior observed at  $\lambda = 0$ . The chosen value of  $\lambda = 10^{-3}$  thus represents a principled balance between reconstruction fidelity and posterior regularization.

Table 2 reports three complementary OOD detection metrics across the  $\lambda$  ablation, each defined as follows. The *uncertainty ratio* is the mean bottleneck uncertainty of anomalous inputs (digit “7”) divided by the mean bottleneck uncertainty of in-distribution

inputs (digit “0”). Bottleneck uncertainty is the sum of predicted variances across the six latent dimensions; a ratio substantially greater than 1 indicates effective OOD discrimination. The *separation percentage* is the fraction of test samples for which the bottleneck uncertainty score correctly separates the two classes when using the midpoint between the class-conditional mean uncertainties as a decision threshold. The *MSE ratio* is the mean reconstruction MSE on anomalous inputs divided by that on in-distribution inputs, providing a complementary reconstruction-quality signal that is independent of the uncertainty mechanism.

**Table 2:** KL Weight Ablation: OOD Detection Performance (Study C). Five independent runs per setting, 200 test samples per class. Normal digit: 0, anomaly digit: 7.

| $\lambda$ | ROC-AUC         | Unc. Ratio         | Separation %      | MSE Ratio       |
|-----------|-----------------|--------------------|-------------------|-----------------|
| 0         | $0.50 \pm 0.22$ | $144.2 \pm 145.3x$ | $44.6 \pm 23.1\%$ | $4.11 \pm 0.23$ |
| $10^{-4}$ | $0.82 \pm 0.09$ | $3.4 \pm 1.2x$     | $13.1 \pm 8.7\%$  | $4.23 \pm 0.25$ |
| $10^{-3}$ | $0.92 \pm 0.03$ | $4.6 \pm 2.7x$     | $26.2 \pm 7.4\%$  | $3.98 \pm 0.42$ |
| $10^{-2}$ | $0.85 \pm 0.04$ | $3.0 \pm 1.1x$     | $27.0 \pm 18.2\%$ | $4.14 \pm 0.33$ |
| $10^{-1}$ | $0.87 \pm 0.12$ | $1.9 \pm 1.1x$     | $16.1 \pm 15.6\%$ | $3.26 \pm 0.35$ |

#### 4.4 Comparative Evaluation Against Baselines

To contextualize the uncertainty quantification capabilities of the SVGP-KAN framework, we conduct comparative evaluations against three established baseline methods: (1) an MLP bottleneck with learned heteroscedastic output, (2) Monte Carlo dropout [16], and (3) deep ensembles [18]. To ensure fair comparison, all methods share identical convolutional encoder and decoder architectures; they differ only in the bottleneck uncertainty mechanism, thereby isolating the contribution of the SVGP-KAN GP layer from any confounds introduced by backbone differences. All methods are trained with the same optimizer, learning rate, and batch size, with training duration matched per study, ensuring an equal training budget. Please see Section 3.2 for further details. To assess the role of model capacity independently of architectural differences, we evaluate two MLP configurations: an architecture-matched variant (thin bottleneck, matching the SVGP-KAN layer structure) and a parameter-matched variant. All methods output both predictive mean and variance and are trained with the Gaussian negative log-likelihood loss to learn heteroscedastic noise. For Monte Carlo dropout, total uncertainty combines the learned aleatoric variance with epistemic uncertainty from inter-pass disagreement across 20 stochastic forward passes. For deep ensembles, total uncertainty combines per-member learned variance with inter-model disagreement across 5 independently trained members.

We do not include a deterministic KAN bottleneck as a baseline, as it produces no predictive uncertainty and therefore cannot be evaluated on the NLL, coverage, or OOD metrics that are the focus of this study; the SVGP-KAN versus MLP bottleneck comparison serves as the appropriate ablation of the GP inference mechanism.

Table 3 reports calibration metrics on the Study A heteroscedastic noise task. SVGP-KAN achieves the lowest negative log-likelihood (-2.05) and RMSE (0.089) among all methods, with  $2\sigma$  coverage of 96.2% against the nominal 95.0%. Both the architecture-matched MLP with fewer parameters (15,154 vs. 27,410) and the parameter-matched MLP (27,514 parameters) achieve worse NLL, indicating that the SVGP-KAN advantage derives from the GP mechanism rather than additional model capacity. Monte Carlo dropout performs comparably to the MLP baselines, consistent with the expectation that dropout-based epistemic uncertainty provides limited additional signal on this well-specified problem. The deep ensemble, while achieving the highest error-uncertainty correlation ( $\rho = 0.71$ ), exhibits the worst NLL (-1.72) and RMSE (0.157), suggesting that individual ensemble members did not fully converge, leading to over-conservative uncertainty estimates (97.3% coverage).

**Table 3:** Study A: Heteroscedastic Calibration Comparison. Nominal  $2\sigma$  coverage: 95.0%. All methods trained with Gaussian NLL.

| Method            | NLL   | RMSE  | $\rho$ | Cov. $2\sigma$ | Params |
|-------------------|-------|-------|--------|----------------|--------|
| SVGP-KAN (Ours)   | -2.05 | 0.089 | 0.45   | 96.2%          | 27,410 |
| MLP (Arch-Match)  | -1.96 | 0.096 | 0.42   | 95.9%          | 15,154 |
| MLP (Param-Match) | -1.97 | 0.095 | 0.41   | 95.8%          | 27,514 |
| MC Dropout        | -1.97 | 0.096 | 0.41   | 95.9%          | 15,426 |
| Deep Ensemble     | -1.72 | 0.157 | 0.71   | 97.3%          | 75,770 |

Table 4 reports OOD detection performance on the Study C anomaly detection task. SVGP-KAN achieves ROC-AUC of 0.94 with an uncertainty ratio of 10.7x between anomalous and normal inputs, substantially outperforming VAE-based approaches (ROC-AUC 0.78–0.82) and Monte Carlo dropout (ROC-AUC 0.73). Deep ensembles achieve the highest ROC-AUC of 0.99, benefiting from reconstruction disagreement across 5 independently trained members. However, this performance requires 3.6 times the parameters (292,995 vs. 82,017). Notably, the parameter-matched VAE (81,981 parameters, matching SVGP-KAN) achieves only 0.78 ROC-AUC, confirming that the GP-based variance mechanism, specifically the orthogonal variance component that naturally increases for inputs distant from inducing points, provides a fundamentally more effective OOD detection signal than learned latent variances with equivalent capacity. The slight advantage of deep ensembles on ROC-AUC reflects a known property of GP-based

models: the posterior variance is bounded above by the prior variance  $\sigma_f^2$ , providing calibrated but finite uncertainty growth for OOD inputs. Ensembles, by contrast, can produce unbounded disagreement for OOD inputs as independently trained members extrapolate in different directions. This trade-off between calibrated bounded uncertainty (SVGP-KAN) and unbounded but potentially uncalibrated disagreement (ensembles) suggests that hybrid approaches combining function-space priors with ensemble diversity may offer further improvements in OOD-sensitive applications.

**Table 4:** Study C: Out-of-Distribution Detection Comparison

| Method            | ROC-AUC | Unc. Ratio | Sep. % | Params  |
|-------------------|---------|------------|--------|---------|
| SVGP-KAN (Ours)   | 0.945   | 10.7×      | 55.5%  | 82,017  |
| VAE (Arch-Match)  | 0.823   | 6.4×       | 3.0%   | 58,989  |
| VAE (Param-Match) | 0.779   | 28.6×      | 7.5%   | 81,981  |
| MC Dropout        | 0.730   | 1.5×       | 3.5%   | 62,759  |
| Deep Ensemble     | 0.994   | 4.9×       | 96.5%  | 292,995 |

Table 5 summarizes the computational cost of each method, measured on a 64×64 grid with batch size 16. We emphasize that the present SVGP-KAN library has not been optimized at all for computational efficiency. Training time per epoch is averaged over 50 epochs after a 5-epoch warmup; inference latency is averaged over 100 forward passes. SVGP-KAN training requires approximately 3.6x the wall-clock time of the architecture-matched MLP baseline per epoch (256 vs. 72 ms), attributable to the moment-matching computations in the GP bottleneck layer. The parameter-matched MLP (27,514 parameters, comparable to SVGP-KAN’s 27,410) incurs only modest additional cost over the thin MLP (80 vs. 72 ms/epoch), confirming that the SVGP-KAN overhead arises from GP inference operations rather than model size. SVGP-KAN inference requires only a single forward pass (46 ms), whereas Monte Carlo dropout requires 20 stochastic passes (360 ms, a 7.8× overhead) and deep ensembles require 5 member evaluations (75 ms). Thus, while SVGP-KAN has higher per-epoch training cost, it is more efficient than both MC dropout and deep ensembles at inference time.

**Table 5:** Computational Cost (Study A architecture, 64×64 Grid, CPU). Batch size 16, averaged over 50 training epochs and 100 inference passes.

| Method            | Parameters | Train (ms/epoch) | Inference (ms/batch) |
|-------------------|------------|------------------|----------------------|
| SVGP-KAN (Ours)   | 27,410     | 256 ± 32         | 46 ± 9 (1× fwd)      |
| MLP (Arch-Match)  | 15,154     | 72 ± 12          | 17 ± 8 (1× fwd)      |
| MLP (Param-Match) | 27,514     | 80 ± 12          | 19 ± 1 (1× fwd)      |
| MC Dropout        | 15,426     | 74 ± 13          | 360 ± 31 (20× fwd)   |
| Deep Ensemble     | 75,770     | 345 ± 9 (5×)     | 75 ± 16 (5× fwd)     |

## 5. Discussion

The three experimental studies are used to examine uncertainty quantification capabilities of our SVGP KAN across different application domains. First, we examine the framework’s capacity for calibrating spatially varying aleatoric uncertainty in scalar field prediction with amplitude-dependent measurement noise. The moderate correlation ( $\rho = 0.45$ ) appropriately reflects irreducible aleatoric noise rather than deterministic prediction, validating that the model has learned the heteroscedastic noise structure — specifically, that observation noise scales with local field magnitude — without overfitting to individual noise realizations.

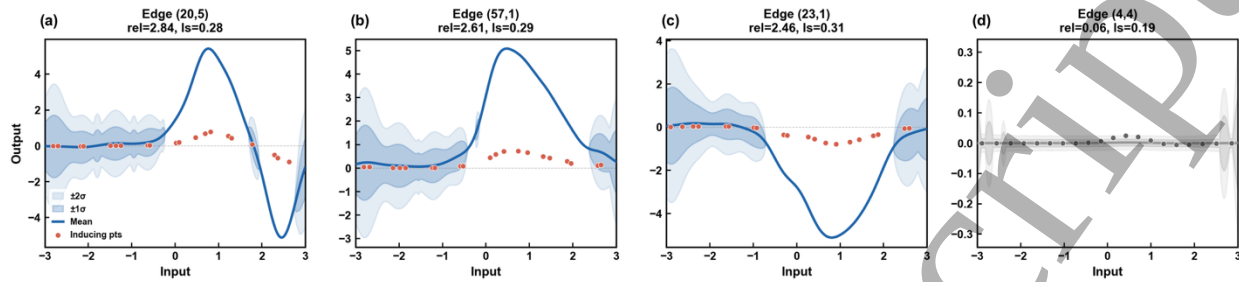
Second, we quantify epistemic uncertainty growth in multi-step forecasting of advection-diffusion dynamics, suggesting that the model correctly identifies degrading prediction reliability over extended forecast horizons. The observed temporal uncertainty growth and large spatial variation occur despite stable underlying physics, confirming that the framework captures model-induced approximation errors rather than physical chaos. This distinction is critical: the neural network’s imperfect dynamics approximation accumulates errors over sequential predictions, and the uncertainty estimates correctly reflect this degradation. The spatial concentration of uncertainty at flow interfaces provides actionable information for adaptive mesh refinement or targeted sensor deployment in operational settings.

Third, we examine out-of-distribution detection capability through bottleneck uncertainty in convolutional autoencoders. This capability emerges from the orthogonal variance term in the predictive variance decomposition, where that uncertainty reverts to the prior  $\sigma_f^2$  when inputs lie far from the inducing point support. Unlike approaches that require omitting KL regularization to preserve OOD detection signals, the orthogonal variance mechanism provides in theory a geometric guarantee that operates independently of the regularization weight  $\lambda$ . This enables a unified framework where all three studies employ principled Bayesian inference, achieving both well-calibrated in-distribution uncertainty (Studies A and B) and reliable out-of-distribution detection (Study C) without task-specific hyperparameter adjustments.

While the three studies presented here employ relatively simple synthetic datasets to facilitate quantitative validation, the SVGP-KAN framework has also been applied to practical scientific data. A recent study [35] reported an uncertainty-aware full-field reconstruction of flow fields from non-time-resolved particle image velocimetry (PIV) data. The framework quantifies spatially varying uncertainty in fully time-resolved flow fields inferred from temporally sparse PIV data.

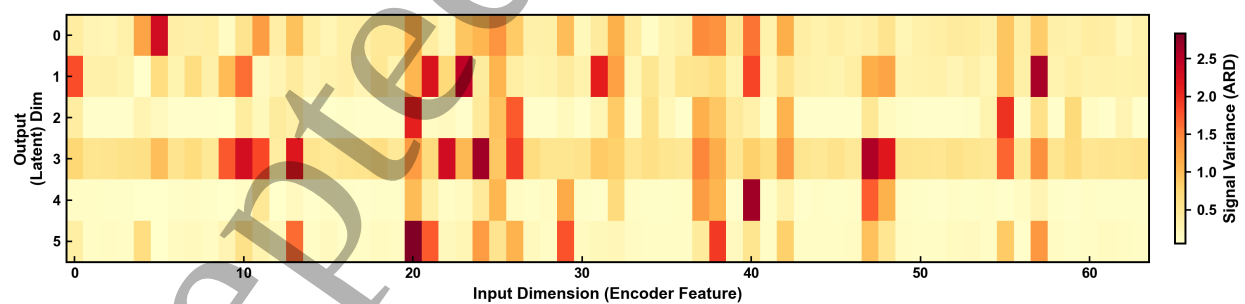
The computational scalability enabled by sparse variational approximations potentially allows application of our framework to large datasets while maintaining  $O(NM^2)$  complexity with  $M \ll N$ . This represents a substantial potential improvement over exact Gaussian process inference ( $O(N^3)$ ). The additive structure of Kolmogorov-Arnold networks enables independent variational approximations across edges, avoiding the factorial complexity that would arise in general deep Gaussian process architectures.

The Kolmogorov-Arnold topology provides three forms of interpretability that are unavailable in conventional multi-layer perceptron architectures. First, each input-output connection is a full univariate Gaussian process rather than a scalar weight, making the learned transformation directly inspectable. Figure 5 contrasts representative high-relevance edge functions with a low-relevance edge from the Study C bottleneck layer (64  $\rightarrow$  6). The high-relevance edges (a–c) exhibit structured nonlinear transformations with tight  $\pm 2\sigma$  confidence bands near the inducing points, indicating data-supported functional relationships. The low-relevance edge (d) shows a near-zero mean function with prior-level uncertainty throughout, indicating that the model has learned to effectively suppress this connection. In a standard MLP, each of these 384 connections would be a single scalar weight offering no indication of what nonlinear transformation is applied or where that transformation is trustworthy. Second, each edge function carries spatially resolved epistemic uncertainty that reveals where the model has data support and where it is extrapolating. Within the inducing point range, posterior variance drops sharply at each inducing location; outside this range, uncertainty reverts to the GP prior. This per-edge uncertainty is intrinsic to the model’s posterior and requires no *post-hoc* analysis.



**Figure 5: Edge Functions: Active vs. Suppressed:** Representative edge functions from the Study C bottleneck (64→6). Panels (a)–(c): high-relevance edges showing structured nonlinear transformations with tight confidence bands near inducing points (red dots). Panel (d): low-relevance edge showing near-zero mean and prior-level uncertainty, indicating effective connection suppression through ARD. Blue lines and bands: posterior mean with  $\pm 1\sigma$  and  $\pm 2\sigma$  confidence intervals.

Third, the automatic relevance determination (ARD) mechanism provides a principled feature relevance map (Figure 6). The learned kernel signal variances across all 384 edges span a 48-fold range, producing effective sparsity through differential weighting: each latent dimension draws on a distinct subset of encoder features, as visible in the structured connectivity pattern of the heatmap. Unlike *post-hoc* attribution methods such as SHAP (SHapley Additive exPlanations) or integrated gradients, this relevance structure is a direct consequence of variational inference and is available at no additional computational cost.



**Figure 6: Edge Relevance Heatmap:** Signal variance (ARD) for all 384 edges in the 64×6 bottleneck layer. Each row corresponds to one of 6 latent dimensions; each column to one of 64 encoder features. Dark entries indicate strongly active connections; pale entries indicate effectively suppressed edges. The structured, non-uniform pattern reveals that each latent dimension draws on a distinct subset of input features.

These three interpretable outputs may serve distinct roles in scientific workflows. First, inspecting learned edge function shapes supports hypothesis generation: a monotone sigmoid relationship between an input and a latent dimension may suggest a threshold nonlinearity worth investigating physically, while a non-monotone shape may indicate competing effects that motivate further experimentation. In PIV flow field reconstruction, for example, edge functions connecting encoder features to latent flow variables may reveal localized sensitivity to seeding-density gradients, suggesting a physically motivated preprocessing step that improves reconstruction fidelity. Second, the ARD relevance map (Figure 6) provides automatic feature selection at no additional computational cost: features whose signal variance is near zero across all latent dimensions can be pruned from the encoder without loss of predictive performance, directly reducing model complexity and highlighting which input channels carry information relevant to the latent state. Third, the per-edge uncertainty bands serve as deployment diagnostics: when uncertainty on a specific edge reverts toward the prior, indicating inputs outside the training support, one may be able to identify which input dimensions are driving unreliable predictions and target data collection or model retraining accordingly.

Several limitations of the present implementation of SVGP-KAN merit acknowledgment. First, the squared exponential kernel employed throughout assumes smooth functions; applications involving discontinuities or sharp transitions may benefit from Matérn kernels or piecewise-defined alternatives. Second, the inducing point count represents a trade-off between approximation quality and computational cost; automatic relevance determination schemes could adaptively allocate inducing points based on local function complexity. Third, the validity and also the theoretical and practical limits of the mean-field approximation remain subjects of rigorous future studies. Fourth, the framework currently assumes Gaussian likelihoods; extensions to Poisson, Bernoulli, or heavy-tailed observation models would broaden applicability to count data, classification, and robust regression. Fifth, while we demonstrate uncertainty propagation through feedforward architectures, extensions to recurrent structures for temporal dynamics or graph neural networks for relational data are also promising directions. Sixth, while the orthogonal variance correction ensures geometric consistency of uncertainty estimates independently of the KL weight, the choice of  $\lambda$  still influences the balance between data fidelity and prior regularization; future work could investigate automatic adaptation schemes that dynamically adjust  $\lambda$  during training based on validation set calibration metrics. Finally, while the three studies employ simple synthetic data with known ground truth to facilitate quantitative validation, more extensive demonstration on larger-scale real-world problems with more complex geometries and physics remains an important direction.

## 6. Conclusion

We have presented a framework for integrating sparse variational Gaussian process inference with Kolmogorov-Arnold network topology, with potential for enabling scalable

Bayesian inference with  $O(NM^2)$  training complexity while maintaining interpretability through additive edge-function structures. The predictive variance decomposition into projected and orthogonal components ensures both well-calibrated aleatoric uncertainty quantification for in-distribution predictions and principled epistemic uncertainty estimation that correctly reverts to the prior for out-of-distribution inputs.

Three experimental studies provide a preliminary examination of the framework’s capabilities across complementary uncertainty quantification tasks. On the heteroscedastic calibration task, the framework delivers lower negative log-likelihood than the MLP bottleneck, Monte Carlo dropout, and deep ensemble baselines under both architecture-matched and parameter-matched conditions. Study B examines epistemic uncertainty growth in multi-step forecasting of stable advection-diffusion dynamics, correctly attributing prediction degradation to model approximation error rather than physical instability. Study C examines out-of-distribution detection through the orthogonal variance mechanism, achieving ROC-AUC competitive with deep ensembles while using fewer parameters and requiring only a single training run, and outperforming VAE-based and Monte Carlo dropout approaches. The advantage of deep ensembles on OOD detection reflects the bounded nature of GP posterior variance, which provides calibrated uncertainty that reverts to the prior rather than the unbounded disagreement possible in ensemble methods. These three examples, however, are rather limited in size, complexity, and scope. Evaluation on larger-scale real-world benchmarks remains an important direction for future work.

We discussed several limitations of the present work. Future work should address these limitations through, for example, extensions to multi-output regression, non-Gaussian likelihoods, and recurrent architectures for temporal dynamics. Automatic adaptation of the KL weight  $\lambda$  based on validation calibration metrics, and integration with physics-informed loss functions, represent promising directions for enhancing the framework’s applicability across scientific disciplines.

## Data Availability

The SVGP-KAN code library is available in <https://github.com/sungjuGit/svgp-kan>.

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